Copies, Conflicts and Broken Parses: How Corpus Defects Affect Retrieval-Augmented Claim Verification

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Abstract

1 Introduction

Retrieval-augmented generation (RAG; Lewis et al., 2020) answers a question from documents retrieved out of a collection, so its answers can be no better than that collection. Real collections are imperfect: documents are duplicated across sources, older versions sit next to newer ones that contradict them, PDF parsing truncates or scrambles text, and chunking splits documents into pieces of varying size. Robustness studies of RAG usually vary what reaches the reader, for instance by adding irrelevant or counterfactual passages to its context (Cuconasu et al., 2024; Chen et al., 2024; Xie et al., 2024), rather than the collection the retriever searches. Practitioners, however, control the collection, and need to know which of its defects are worth cleaning.

We ask how much each of four common defects, on its own, changes retrieval and the final answer. We use claim verification because it has labels for both the answer (does the evidence support or contradict the claim?) and the evidence (which documents), so a defect's effect can be traced from retrieval to the verdict. Defects are injected under control, one type and rate at a time, into an otherwise unchanged corpus, with every model, prompt and threshold held fixed, so any change in the results comes from the corpus. Our contributions:

- A seeded corruption tool for four defect types (near-duplicates, stale copies with flipped findings, broken parsing, chunking) that logs every change, and a pipeline that regenerates and verifies each corrupted corpus from its seed.
- Paired, claim-level measurements on SciFact (Wadden et al., 2020) for two retrievers, a reranker and four verifiers, repeated over five corruption seeds and three verifier training seeds, and replicated on Climate-FEVER (Diggelmann et al., 2020).
- Evidence that retrieval metrics rank the defects in the wrong order of harm, that stale copies matter only to verifiers that read their context closely, and that small chunks make a fine-tuned verifier confidently wrong.

All code, per-claim predictions, corruption manifests and fine-tuned models are public.[1]

2 Related Work

Robustness of retrieval-augmented models. Language models are easily distracted by irrelevant context (Shi et al., 2023), use the middle of long contexts poorly (Liu et al., 2024), and can be trained to ignore irrelevant passages (Yoran et al., 2024). RGB (Chen et al., 2024) tests noise robustness, negative rejection and counterfactual robustness of RAG readers, and Cuconasu et al. (2024) find that random documents in the context can even help. These studies manipulate the reader's context directly; we manipulate the collection and let the retriever decide what reaches the reader.

Knowledge conflicts. Longpre et al. (2021) substitute entities in passages to create conflicts with a model's parametric knowledge, and Xie et al. (2024) show that LLMs readily accept coherent conflicting evidence. Our stale copies create conflicts *within* the collection: the original and a flipped copy are both retrievable, as when outdated and current versions of a document coexist.

Duplicates and data quality. Near-duplicates are pervasive in web-scale text and deduplica-

[1] https://github.com/Likhithvenkatanalla/rag-dq-benchmark; archived at https://doi.org/10.5281/zenodo.23314458.

tion improves language-model training (Lee et al., 2022). Retrieval benchmarks such as BEIR (Thakur et al., 2021) evaluate retrievers across clean collections; we hold the task fixed and vary the collection's quality.

Claim verification. FEVER (Thorne et al., 2018), SciFact (Wadden et al., 2020) and Climate-FEVER (Diggelmann et al., 2020) pair claims with evidence and a verdict, which makes the path from evidence to answer measurable.

3 Setup

3.1 Data

SciFact pairs expert-written scientific claims with 5,183 abstracts. We evaluate on the 300 dev claims (124 SUPPORT, 64 CONTRADICT, 112 NEI); 188 have gold evidence, among 184 distinct gold abstracts. The 809 train claims, disjoint from dev, are used only for training and threshold tuning.

Climate-FEVER pairs real-world climate claims with five Wikipedia sentences each, labelled by annotators. We build a closed corpus from all 5,240 distinct evidence sentences (each a record titled with its article), close in size to SciFact, and drop the 154 DISPUTED claims, leaving 1,381 claims (654 SUPPORT, 253 CONTRADICT, 474 NEI). A claim's gold records are its evidence sentences labelled with the claim's own verdict (907 claims have at least one). No model is trained on Climate-FEVER.

3.2 Pipeline

Retrieval. BM25 (Robertson and Zaragoza, 2009) over whitespace tokens, and dense retrieval with BGE-small-en-v1.5 (Xiao et al., 2023) and exact cosine search (Johnson et al., 2019). Recall@k is the share of a claim's gold records found in the top k, averaged over claims with evidence.

Verification. A verifier reads the claim with the top-3 dense-retrieved records and outputs SUPPORT, CONTRADICT or NEI. *Qwen-1.5B* and *Qwen-3B* are Qwen2.5-1.5B/3B-Instruct (Qwen Team, 2024), zero-shot with greedy decoding, instructed to answer with one word. *NLI zero-shot* is the cross-encoder `nli-deberta-v3-xsmall` (He et al., 2023) as released, and *NLI fine-tuned* is the same model fine-tuned for 3 epochs on 1,123 SciFact train pairs (gold abstracts with their labels; cited abstracts of claims without evidence and the top non-gold BM25 abstract of each claim with evidence as NEI), trained with three seeds. An NLI verifier scores each (claim, record) pair; the claim takes the higher of the maximum SUPPORT and CONTRADICT probabilities over its three records, or NEI if neither reaches a threshold tuned for macro-F1 on 20% of the train claims held out from training (never on dev).

Reranking. A MiniLM cross-encoder (Nogueira and Cho, 2019), as released or fine-tuned on SciFact train, reorders the dense top-10 before verification.

3.3 Defects

Each defect is applied to a share (10, 25 or 50%) of the gold-evidence records, so that it reaches the evaluated claims; claims and labels never change.

- **Duplicate** adds a near-copy of the record (one sentence dropped if there are more than two, about 5% of words deleted, one adjacent word pair swapped per sentence).
- **Stale** adds a copy whose directional findings are flipped by rule (*increased*↔*decreased*, *higher*↔*lower*, "is associated" → "is not associated", …; for Climate-FEVER also *warming*↔*cooling*, *rise*↔*fall*, …). Copies with nothing to flip are identical to the original; we analyse flipped copies separately.
- **Parse** replaces the record with a broken parse: truncated at 30–60% of its length, sentences scrambled, or OCR-style character noise.
- **Chunk** splits every SciFact abstract into records of 1, 3 or 5 sentences (not applicable to one-sentence Climate-FEVER records).

A copy is a different document: retrieving it does not count as retrieving the gold record. A chunk counts as its parent abstract.

3.4 Statistics

All runs on a dataset share the same claims, so comparisons are paired: for each claim we record whether the answer went from right to wrong (*lost*) or wrong to right (*gained*) relative to the clean corpus, and report both counts with an exact McNemar test (McNemar, 1947). We run many such tests without correction, so we treat as findings only effects whose direction repeats across corruption seeds and, for the fine-tuned verifier, training seeds.

Corpus	BM25	Dense	+ reranker
clean	0.615	0.744	0.805
duplicate 10/25/50%	.594/.541/.463	.689/.668/.604	.774/.751/.644
stale 10/25/50%	.583/.530/.466	.691/.622/.551	.752/.657/.571
parse 10/25/50%	.615/.599/.578	.744/.734/.739	.795/.799/.758
chunk 5/3/1 sent.	.583/.590/.520	.749/.742/.762	.803/.818/.784

Table 1: SciFact recall@1 (188 claims with evidence; corruption seed 0). Recall@10 is unchanged by copies (0.883 BM25, 0.957 dense).

4 Results on SciFact

4.1 Clean baseline

Dense retrieval beats BM25 at every cut-off (recall@1 0.744 vs. 0.615; recall@10 0.957 vs. 0.883). Qwen-1.5B reaches 0.557 accuracy and 0.502 macro-F1 (majority class: 0.413 and 0.195) but answers SUPPORT for 200 of 300 claims. The fine-tuned NLI verifier reaches 0.650 / 0.680 / 0.667 for its three training seeds, each significantly above Qwen-1.5B (lost/gained 32/60, 36/73, 35/68; $p \leq 0.005$). Untuned, the NLI model reaches only 0.353.

4.2 Retrieval

Copies push the gold abstract out of first place in proportion to their rate (Table 1, Figure 1) while recall@10 does not change: a copy usually sits just above its original. The fine-tuned reranker lifts clean recall@1 to 0.805 and recovers part of the duplicate loss but little of the stale loss: a flipped copy is as relevant to the claim as its original, so a relevance model cannot tell them apart. Broken parsing removes documents from BM25's reach (recall@10 0.845 at 50%) but barely affects dense retrieval, and chunking hurts BM25 but not dense retrieval.

4.3 Verification

4.4 Conflicting evidence

4.5 Calibration

5 Replication on Climate-FEVER

6 Discussion

Limitations

Ethics and Reproducibility Statement

SciFact and Climate-FEVER are public research datasets; they are downloaded from their official releases and not redistributed. All experiments ran on free CPU runners of GitHub Actions; the workflows that produced every number are in the repository, and each corrupted corpus is regenerated from its seed and checked against the committed manifest. Code was developed with AI assistance (Claude); every reported number comes from runs whose outputs are in the repository.

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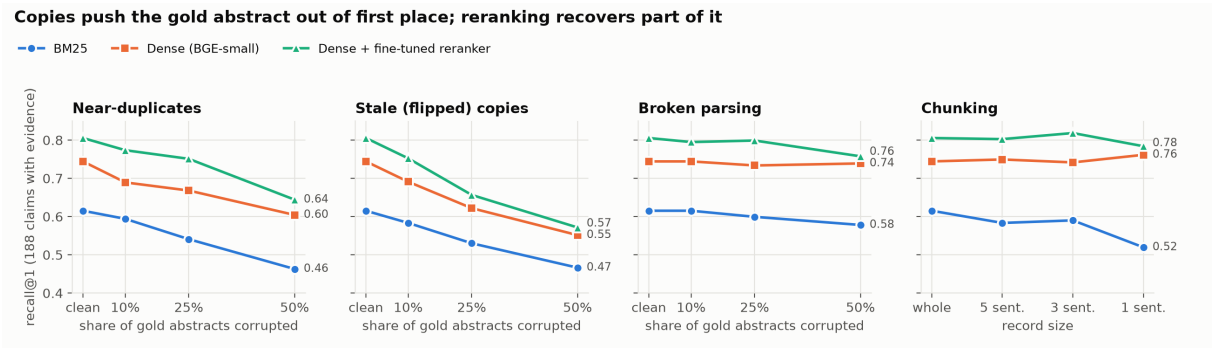


Figure 1: SciFact recall@1 by corruption level for BM25, dense retrieval and dense retrieval with the fine-tuned reranker (corruption seed 0).

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